

# JIACHANG ZHANG

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Date of Birth: 11st, March, 2001 Gender: Female

## EDUCATION

**Nanjing University**, Nanjing, China Sep 2022 - June 2027 (expected)
School of Astronomy & Space Science, Ph.D. candidate, Astrophysics
Supervisor: Prof. Zhiyuan Li
Thesis: X-ray Variability of Accreting Compact Objects in Nearby Galaxies

**Nanjing University**, Nanjing, China Sep 2018 - June 2022
Kuangyaming Honors School, B.S., Astronomy

## RESEARCH INTERESTS

- **Time-domain studies of X-ray binaries and accreting compact objects;**
- **Compact-object populations and their stellar environment in nearby galaxies;**
- **X-ray source populations in star clusters;**
- **Diffuse ionized gas and stellar feedback.**

## ACADEMIC VISIT

**University of Washington**, Seattle, WA, US
Sep 2024 - Dec 2024
Supervisor: Prof. Benjamin F. Williams
Ionized gas and related star populations in M31 and M33

## SELECTED PRESENTATIONS

**46th COSPAR Scientific Assembly**, Florence, Italy, Aug 2026.
"Beyond the Milky Way: Systematic X-ray Timing of Extragalactic LMXBs in M31, M81 and Cen A with the Chandra Archive" *(Oral Presentation)*

**Frontiers in Diffuse Hot Gas and Space-based X-ray Telescopes**, Suzhou, Jiangsu, China, Jun 2026.
"Recurrent Luminous X-ray Flares from a Persistent X-ray Binary in M81" *(Oral Presentation)*

**The 2025 Annual Science Meeting and Early Science Workshop of the Chinese Space Station Survey Telescope (CSST)**, Kunming, Yunnan, China, Jul 2025.
"A High-Resolution Study of Hα-emitting Sources in M33" *(Oral Presentation)*

**The 2025 Workshop on the Structure, Formation, and Evolution of Stellar Clusters**, Dali, Yunnan, China, Mar–Apr 2025.
"X-ray Emission from Star Clusters in M31 and M33" *(Oral Presentation)*

**25 Years of Chandra Science Symposium**, Boston, Massachusetts, USA, Dec 2024.
"A *Chandra* Search for Periodic X-ray Sources in the Bulge of M31" *(Poster Presentation)*

## PROPOSAL EXPERIENCE

**Awarded Observing Program**

**2024 — Co-Investigator**, "How Do Stellar Feedback WEAVE into the ISM: the Unique Case of M31," WHT/WEAVE Open-Time Program, 4.0 hr awarded (PI: Matthew Smith).

**China Space Station Telescope (CSST) Early Science Program**

**Co-Investigator**, "Joint MCI–IFS Observations of M31".

**Co-Investigator**, "Joint IFS–MCI Observations of the Virgo Cluster".

**Co-Investigator**, "Joint MCI–IFS Observations of M33".

## MAIN AWARDS AND HONORS

**2024** — Tiangong International Exchange Scholarship, Nanjing University (≈USD 10,000)

**2024** — Outstanding Poster Award, The First Shen Kuo Symposium of the Chinese Astronomical Society, Wuhan, Hubei, China *"A Chandra Search for Periodic X-ray Sources in the Bulge of M31"*

**2022** — Lu Dexin Award for Outstanding Graduates, Nanjing University

**2020** — Meritorious Winner, Mathematical Contest in Modeling (MCM), COMAP

**2019, 2020** — People's Scholarship, Nanjing University

**2019, 2020** — Scholarship for Top-Talent Students, Nanjing University

## TEACHING AND MENTORING

- **Teaching Assistant**

Course: ***An Introduction to X-ray Astronomy,*** *2026 Spring Semester,* Nanjing University

— X-ray data processing training project, teaching students how to reduce the X-ray data, analyze the spectrum and light curves.

- **Student Mentoring**

**2025-present —** Yuyan Wang, Undergraduate Innovation and Entrepreneurship Training Program, "*An X-ray Study of Massive Stars in the Solar Neighborhood*" — Mentored a project using archival Chandra and Gaia data to identify and characterize long-duration stellar X-ray flares and investigate their statistical properties.

**2026-present —** Yuchi Zhang, Undergraduate Research, "*Spectral Evolution and State Variability of X-ray Binaries in M31*" — Supervising a project using archival X-ray observations to investigate spectral evolution, luminosity-dependent behavior, and possible accretion-state transitions in M31 X-ray binaries.

## SKILLS

**X-ray Data Analysis:** Extensive experience in the reduction and analysis of data from **Chandra, XMM-Newton, Swift, and NuSTAR**, including source detection, spectral extraction and modelling, light-curve analysis, periodic and aperiodic timing analysis, and searches for transient and variable sources.

**Optical Imaging Analysis:** Experience with **Hubble Space Telescope (HST)** imaging for point-source photometry and the analysis of diffuse ionized gas and emission-line sources; participant in the early-science program of the **CSST**.

**Astronomical Software:** CIAO, SAS, HEASoft, XSPEC, DS9, DOLPHOT, Astropy and so on.

**Programming and Scientific Computing:** Python, IDL, Bash/Shell, and C++; LaTeX for scientific writing and typesetting.

**Languages:** Mandarin Chinese (native); English (professional working proficiency).

## PUBLICATIONS

**Zhang, J.**, et al. 2026, "The Dancing Bonfire: An Unusual Persistent X-ray Source in M81 Exhibiting Recurrent Luminous Flares," accepted for publication in ApJ, https://arxiv.org/abs/2610.23339

**Zhang, J.**, Li, Z., Hua, Z., & Bao, T. 2026, "Probing periodic and aperiodic variability of X-ray sources in M31, M81, and Centaurus A with Chandra," *MNRAS*, 546, staf2237. doi: 10.1093/mnras/staf2237.

Ma, Q.-Q.^†, **Zhang, J.^†**, Gu, W.-M., Li, Z., Weng, S.-S., & Bao, T. 2026, "The first extragalactic ultra-compact X-ray binary: A candidate black hole-white dwarf system," *The Innovation*, 7, 101181. doi: 10.1016/j.xinn.2025.101181. (Co-first Author)

**Zhang, J.**, Bao, T., & Li, Z. 2024, "A Chandra search for periodic X-ray sources in the bulge of M31," *MNRAS*, 530, 2096–2113. doi: 10.1093/mnras/stae1002.

Li, Z., Li, Z., Luo, G., Liu, S.-Y., Jiao, S., Meng, F., Ren, Z., Iono, D., **Zhang, J.**, Li, D., Tsai, C.-W., & Kim, J. 2026, "NOEMA Observations of the Center of M33: A Parsec-scale Molecular Filament and an Ordered Gas-Star-Formation Sequence," accepted for publication in AJ. https://arxiv.org/abs/2610.35278.

Li, Z., Su, Z., Wang, S., Zhou, Y. F., Li, Z., Lyu, X., Meng, F., Zhang, K., Li, D., Tsai, C., Wu, J., & **Zhang, J.** 2025, "Ring or No Ring—Revisiting the Multiphase Nuclear Environment in M31," *ApJ*, 983, 182. doi: 10.3847/1538-4357/adc103.

## OUTREACH AND ACADEMIC SERVICE

**2023 — Volunteer**, Undergraduate Summer Camp, School of Astronomy and Space Science, Nanjing University

**2024 — Student Speaker**, Undergraduate Summer Camp, School of Astronomy and Space Science, Nanjing University

**2023–2025 — Volunteer**, Astronomy Night Public Outreach Program, Nanjing University